Uniform distribution

1 RECOGNISE IT

- Every value between fixed bounds a and b is equally likely.
- Rectangular/constant density; zero outside [a,b].
- Probability depends only on interval length.

2 DECISION / METHOD

1. Identify support [a,b].
2. Clip the requested interval to the support.
3. Probability = overlap length / total length.
4. Use midpoint for the mean.

3 FORMULA

$f(x)=1/(b-a),\ a<=x<=b$
$P(c<X<d)=(d-c)/(b-a)$
$E(X)=(a+b)/2$
$Var(X)=(b-a)^2/12$

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, expected value and variance.
Still know: interval probability as length ratio and support clipping.

5 EXCEL ROUTE

No special function needed:
`=(upper-lower)/(b-a)`
SD if needed: `=SQRT((b-a)^2/12)`.

6 TINY WORKED EXAMPLE

For X~Uniform(2,10),
$P(5<X<8)=(8-5)/(10-2)=0.375$.

7 TOP EXAM TRAPS

- Saying every interval has equal probability; only equal-length intervals do.
- Using density 1/(b-a) as the final interval probability.
- Including area outside [a,b].

8 SAY ALOUD

Uniform probability is the requested overlap length divided by the full support length.

9 EXIT CHECK

For Uniform(-2,6), what is P(X>4)?

ONE-GLANCE VISUAL ROUTE

